

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY****Sixth Semester of B. Tech. (CL) Examination****April - May, 2018****CL305.01 Environmental Engineering II****Date: 08.05.2018, Tuesday****Time: 10:00 a.m. To 1:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

**SECTION – I****Q - 1 Answer the questions below: [07]****(1) What is Sludge Volume Index? [01]****(2) Drop manhole is used in sewers when\_\_\_\_\_. [01]****(3) Identify various sources of the following air pollutants & also state their effects [03]****on human health:****(i) Sulphur dioxide,(ii) Carbon Monoxide,(iii) Oxides of Nitrogen****(4) Define following terms with respect to leaf damage due to air pollution(Any 2): [02]****(i) Necrosis, (ii) Chlorosis (iii) Epinasty****Q - 2(a) Enlist the control methods for noise pollution by different sources. [02]****(b) Define 'Environmental Pollution'. Discuss the impacts of waste dumping on aquatic life. [06]****(c) Describe principle and working of a settling chamber with neat sketch. [06]****OR****(c) Describe with neat sketch the principle and working of an Electrostatic Precipitator. [06]****Q - 3(a) What is meant by smog? Discuss its causes and effects. [05]****(b) Attempt any three: [09]****(i) Draw a flow chart showing functional elements of municipal solid waste management. What do you understand by House to house collection and Communal storage collection? [03]****(ii) What is a Transfer Station? When would a transfer station be provided? Enlist types of transfer stations employed for Municipal solid waste. [03]****(iii) Discuss various methodology of composting of solid waste. [03]**

- (iv) Discuss parameters governing the choice of sanitary land filling for solid waste. [03]

### SECTION – II

**Q - 4 Do as directed:** [07]

- (1) Why BOD value is less than COD value for a sewage sample? [01]
- (2) In the COD test, a sample is digested with potassium dichromate in the presence of sulphuric acid. Which of these chemicals (Manganus sulfate, silver sulphate, mercuric sulphate) are added respectively, as a catalyst and a preventer of chloride interference? [02]
- (3) What happens to following parameters when wastewater temperature increases? Also briefly mention the reason. (a) Dissolved Oxygen (increase/decreases), (b) Velocity of settling of given particles in this wastewater (increase/decreases). [02]
- (4) What are the effects of following compounds/parameters on disposal of domestic wastewater? [02]  
(a) Biodegradable organic matter (b) Sulphates

- Q - 5(a)** Draw a neat sketch of treatment scheme used in a conventional sewage treatment plant. Give functions of each treatment unit. [05]
- (b)** Draw a graph of BOD v/s incubation period. Also derive an expression for first stage BOD. [05]

### OR

- (b)** The BOD of a sewage incubated for one day at 30°C has been found to be 100 mg/L. What will be the five day 20°C BOD? Assume  $K = 0.12$  (base 10) at 20° C, and  $\theta = 1.056$ . [05]
- (c)** Which type of settling prevails in a primary settling tank used for sewage treatment? Determine the dimensions of rectangular primary settling tank for the average design flow 0.44 m<sup>3</sup>/s. Consider overflow rate = 36 m<sup>3</sup>/m<sup>2</sup>/day. [04]

- Q - 6(a)** Compare the working of screen chamber and grit chamber? Design aerated grit chamber for a flow of 1.321 m<sup>3</sup>/s. Consider supply of air as 5-8 L/s/m. [07]
- (b)** Discuss variation in sewage flow and design periods of different components of sewerage system. Also illustrate with neat sketches the various types of storm water inlets. [07]

**OR**

- (a)** (i) Differentiate between attached and suspended growth processes. **[07]**  
(ii) Explain briefly various methods of aeration employed in activated sludge treatment.
- (b)** Justify use of Combined sewers and also discuss its advantages and disadvantages. **[07]**  
Calculate velocity of flow and discharge in a circular sewer having dia. 1.00 m laid at a gradient 1 in 500. Take  $N = 0.012$  in Manning's formula. The sewer runs at 50% depth.

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